

FRACTALSCAPE: USER GUIDE

FractalScape is an interactive application that allows you to generate 2D landscapes using known fractal algorithms (fractal trees, koch snowflake) and randomness. The terrain itself is generated with perlin noise, which also exhibits fractal-like patterns.

User interface

After running the application, you can adjust the following parameters (using sliders in the application window):

1. Background Color and Terrain Color
 - Hue of background gradient/terrain in the HSB color model
 - For values in range (180, 300), koch snowflakes will be added automatically
2. Hilliness - Flatness
 - Ranges from 1 (hilly terrain) to 60 (flat terrain)
3. Level of Detail
 - Step between individual terrain layers
 - Ranges from 1 (most detailed) to 5 (least detailed)
4. Tree Frequency
 - Ranges from 2 (highest number of trees) to 20 (smallest number of trees)
5. Noise Seed
 - Sets the noise seed for terrain generation
 - Ranges from 0 to 1000

Click Confirm Changes to redraw the sketch with adjusted parameters.

After clicking the Export Image button, the sketch is exported in two formats:

- .jpg (raster format with screen resolution)
- .svg (vector format)

You can save the current parameters via Save Parameters button, or load previously saved parameters via Load Parameters button.

All files are saved into the application folder.